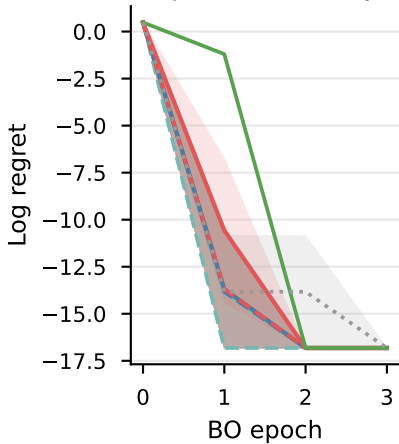
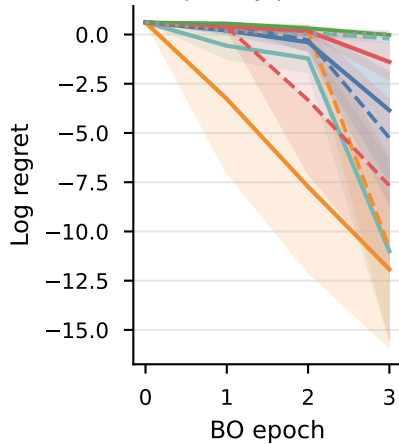


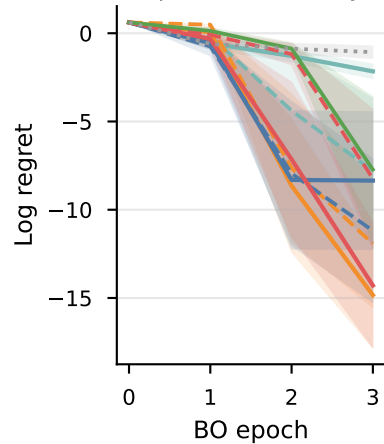
trpB / one to many



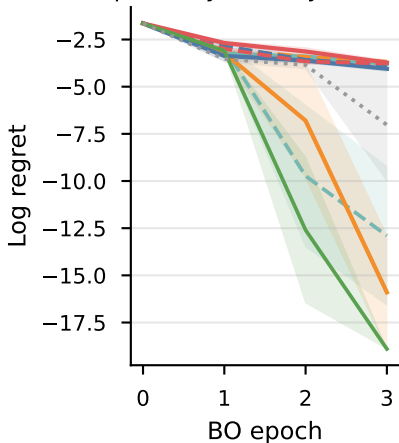
trpB / by position



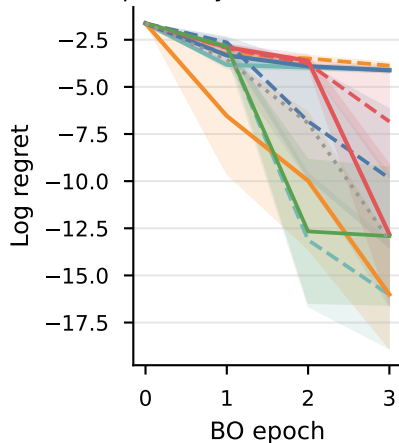
trpB / two to many



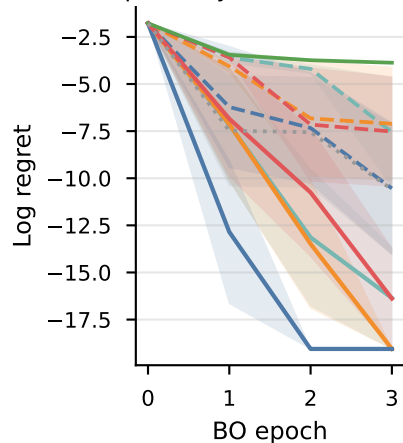
alpha-amylase / by mutation



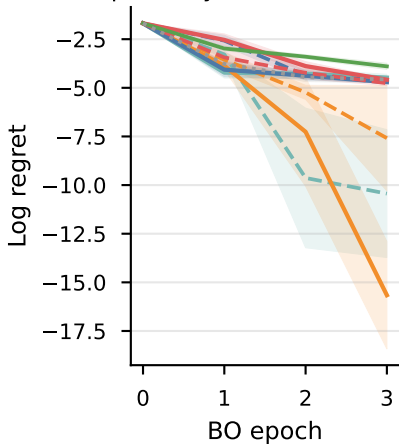
alpha-amylase / close to far



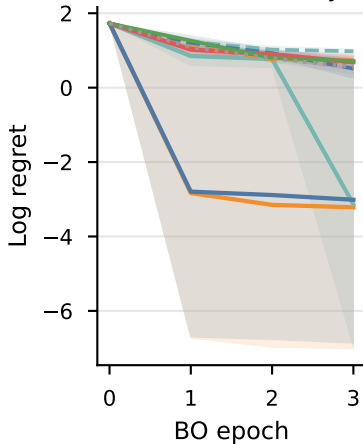
alpha-amylase / far to close



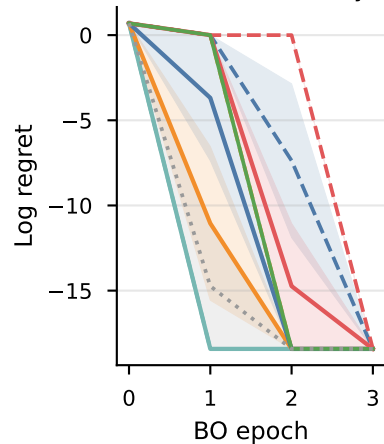
alpha-amylase / one to many



ired / two to many



nucB / two to many



--- T5 frozen --- ProtT5 frozen --- ESM2 frozen --- ESMC frozen --- One-hot one_hot
 --- T5 ft --- ProtT5 ft --- ESM2 ft --- ESMC ft Random random